AI-Powered Customer Support Ticketing System using Salesforce

Name:
Utkarsh Ukey

Platform:
Salesforce Developer Org

Program:
Salesforce Administrator with AI Agentforce

Organization:
SmartBridge

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1. Project Overview

Customer support plays an important role in maintaining customer satisfaction and trust in any organization. Companies receive many support requests every day related to technical issues, product queries, or service complaints. Managing these requests manually can be difficult and time-consuming. Without a proper system, support teams may lose track of requests, delay responses, or fail to respond to customers efficiently.

To address this challenge, the **AI-Powered Customer Support Ticketing System** is developed using the **Salesforce platform**. Salesforce is a cloud-based Customer Relationship Management (CRM) platform that helps organizations manage customer information, automate processes, and improve service operations.

This system centralizes customer support operations by managing customer data through **Account** and **Contact** objects and handling support requests through the **Case** object. In addition to standard objects, the system also includes custom objects such as **Support Category**, **SLA Policy**, and **AI Feedback** to extend system functionality.

A key feature of the system is the integration of **Salesforce Agentforce AI**, which generates **AI-based suggested responses** for customer issues. When a case is created and its details are provided, the AI analyzes the case information and produces a recommended reply that support agents can review and send to customers.

The system also includes reporting and dashboards that allow managers to monitor case activity and system performance.

Overall, this project demonstrates how Salesforce tools and AI capabilities can be used to create a structured and efficient customer support management system.

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2. Problem Statement

Many organizations still manage customer support requests manually or through basic ticket management systems. This approach creates several challenges for both support teams and customers.

One major problem is the lack of proper organization of support tickets. When support requests are not categorized correctly, agents may struggle to understand the issue and determine the best solution. This may result in delays in responding to customers.

Another challenge is that support agents may spend a lot of time writing responses to common customer issues. Without assistance tools, agents must manually analyze the problem and draft replies, which can slow down the support process.

Organizations also face challenges in monitoring support operations. Without clear reports and dashboards, managers cannot easily track case activity or evaluate support performance.

Therefore, there is a need for a system that can:

• Organize and manage customer support requests
• Assist support agents with AI-generated response suggestions
• Maintain centralized customer information
• Provide reporting and monitoring capabilities

The **AI-Powered Customer Support Ticketing System** addresses these challenges by using Salesforce features along with AI-based response generation.

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3. Project Objectives

The main objective of this project is to design and implement an intelligent customer support ticketing system using the Salesforce platform.

One of the key objectives is to centralize customer information so that support agents can easily access customer details and support history. This is achieved using Salesforce standard objects such as **Account** and **Contact**.

Another important objective is to implement **AI-based assistance for support agents**. The system uses **Salesforce Agentforce AI** to analyze case details and generate suggested responses that agents can use when replying to customers.

The project also aims to improve agent productivity by providing AI-generated suggestions directly on the Case record. These suggestions help agents respond to customer issues more quickly and consistently.

Another objective is to implement **role-based access control** using profiles and permission sets so that users only have access to the data and features required for their role.

The system also focuses on performance monitoring by using **Salesforce reports and dashboards** to track case activity and support operations.

Finally, the project aims to provide practical experience with Salesforce administration concepts such as object configuration, automation using flows, security management, and reporting.

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4. Stakeholders

Stakeholders are individuals or groups who interact with the system or are affected by its performance. The **Customer** is one of the primary stakeholders. Customers create support requests when they experience problems with products or services. Their main expectation is to receive timely assistance and solutions. The **Support Agent** is responsible for handling customer cases. Agents review case details, analyze customer issues, and respond using AI-generated suggestions provided by the system. The **Support Manager** monitors the performance of the support team. Managers use reports and dashboards to track case status and evaluate support activities. The **System Administrator** manages the technical configuration of the system. Administrators create objects, configure automation, manage security settings, and maintain the overall system. Each stakeholder interacts with the system differently, and the system is designed to support their specific responsibilities.

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5. System Requirements

Before implementing the system, certain technical requirements must be satisfied.

The main platform used for this project is the **Salesforce Developer Organization**, which provides a free environment for building and testing Salesforce applications.

A web browser such as **Google Chrome** is required to access the Salesforce platform. Since Salesforce is cloud-based, all configuration and development activities are performed through the browser interface.

The system also requires a **computer or laptop with a stable internet connection** to access Salesforce and perform configuration tasks.

In addition, the project requires basic knowledge of Salesforce administration concepts such as object configuration, flows, profiles, permission sets, and reporting.

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6. Data Architecture

Data architecture defines how data is organized and stored in the system.

The **Account object** is used to store information about customer organizations or companies. Each account represents a customer entity and can have multiple associated contacts.

The **Contact object** stores detailed information about individual customers such as their name, email address, and phone number. Contacts are linked to accounts and represent the individuals interacting with the support system.

The **Case object** is the core component of the ticketing system. It represents a customer support request or issue. Each case contains information such as issue description, priority level, status, and assigned support agent.

In addition to these standard objects, the system includes custom objects such as **Support Category**, **SLA Policy**, and **AI Feedback** to support additional functionality and data tracking.

These objects together form the data structure of the customer support ticketing system.

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7. System Architecture

The system architecture explains how different components interact to manage customer support operations.

The process begins when a **customer support request is recorded as a Case in Salesforce**. The case contains information about the issue faced by the customer.

Support agents can then view the case details within Salesforce. Using **Agentforce AI**, the system analyzes the case description and generates a **suggested response** for the support agent.

The agent reviews the AI-generated response, makes any necessary edits, and sends the response to the customer.

Reports and dashboards are used to monitor case activity and provide visibility into support operations.

This structured workflow helps ensure that customer support requests are handled efficiently

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8. Automation Implementation

Automation is used in the system to support support operations and reduce manual effort.

Salesforce **flows and automation tools** are configured to assist with tasks such as sending notifications or triggering actions when cases are updated.

The system also uses **Prompt Builder and Agentforce AI** to automatically generate suggested responses based on case details. These responses assist support agents in resolving customer issues more efficiently.

Automation improves consistency in responses and reduces the time agents spend drafting replies.

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9. Security Model

Security is an important part of the system. In this project, **role-based access control** is implemented using profiles and permission sets.

Profiles define the basic level of access that users have within the system. For example, a **Support Agent profile** may allow users to create and update cases but restrict access to administrative settings.

Permission sets provide additional access to specific users without changing their profiles. This makes the security model flexible and easier to manage.

By using profiles and permission sets, the system ensures that users only access data relevant to their role.

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10. Reporting and Monitoring

Reports and dashboards are used to monitor support operations and analyze case activity.

Managers can generate reports to track case status such as **open, in-progress, and closed cases**.

Agent performance reports can also be created to evaluate how many cases have been handled by support agents.

Dashboards provide visual representations of these reports using charts and graphs. This helps managers quickly understand support performance and identify trends.

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11. AI Feedback Loop

The system includes an **AI feedback mechanism** to support continuous improvement.

Support agents can provide feedback on the usefulness of AI-generated suggested responses. This feedback is stored in the **AI Feedback object**.

Administrators can review this feedback to understand whether the AI suggestions are helpful and identify areas where improvements may be required.

This feedback loop helps improve the effectiveness of AI-assisted support over time.

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12. Testing

Before completing the project, several tests were performed to verify system functionality.

Sample **Accounts and Contacts** were created to represent customers. Test cases were then generated to simulate customer support requests.

The AI response generation feature was tested to ensure that it produced appropriate suggestions based on case descriptions.

Reports and dashboards were also reviewed to ensure that they displayed accurate case information.

Testing helped ensure that the system worked as expected before final documentation.

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13. Maintenance and Troubleshooting

After implementation, the system requires regular maintenance to ensure smooth operation.

Administrators may update flows, modify configurations, or adjust security settings as business requirements change.

Troubleshooting may also be required if automation or AI features do not behave as expected.

Regular monitoring and updates help maintain system performance and reliability.

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14. Learning Outcomes

This project provided valuable hands-on experience with Salesforce administration and AI integration.

During development, several important skills were learned, including **data modeling, automation configuration, security management, and report creation**.

The project also helped demonstrate how **AI tools such as Agentforce can assist support agents by generating suggested responses based on customer issues**.

Overall, the project improved practical knowledge of Salesforce and demonstrated how AI and CRM platforms can improve business processes.

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15. Conclusion

The **Salesforce AI-Powered Customer Support Ticketing System** demonstrates how Salesforce tools and AI capabilities can improve customer support operations. By using standard and custom objects to manage customer data and cases, along with **Agentforce AI for generating suggested responses**, the system assists support agents in resolving issues more efficiently.

The use of reports, dashboards, and structured data management provides better visibility into support activities. The system also incorporates security controls and feedback mechanisms to ensure reliable operation and continuous improvement.

Overall, the project provides practical experience with Salesforce administration and highlights the benefits of integrating AI technologies into customer support systems.

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16. Screenshots

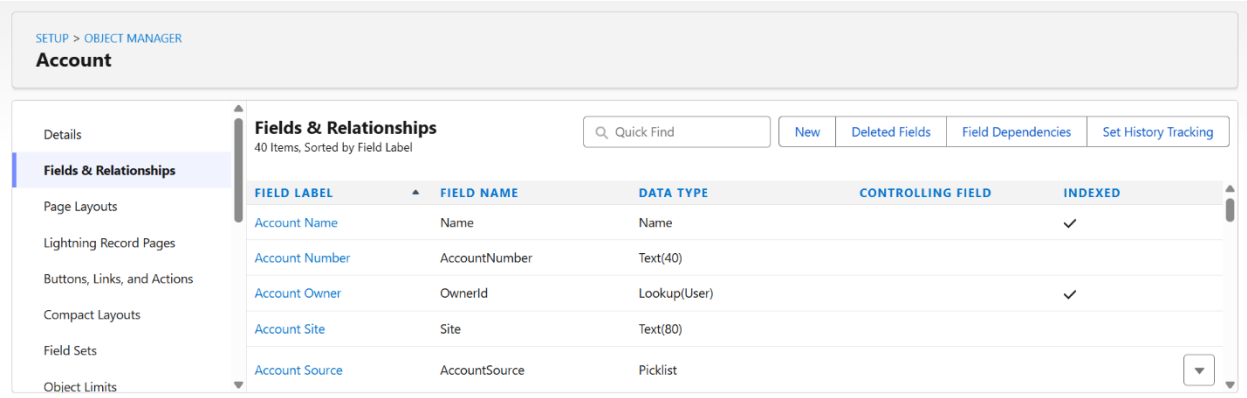


Figure 1: Account Object for Managing Customer Organizations

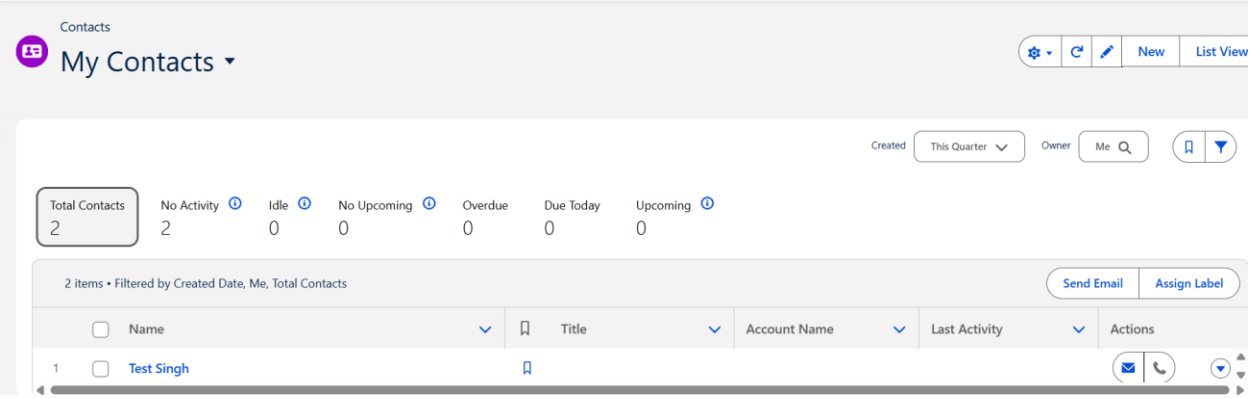


Figure 2: Contact Object for Storing Customer Details

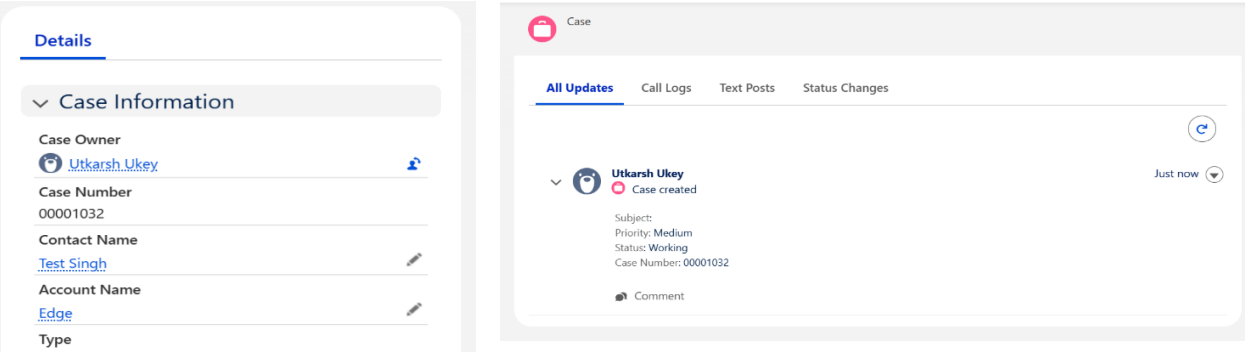


Figure 3: Case Object Used for Managing Customer Support Tickets

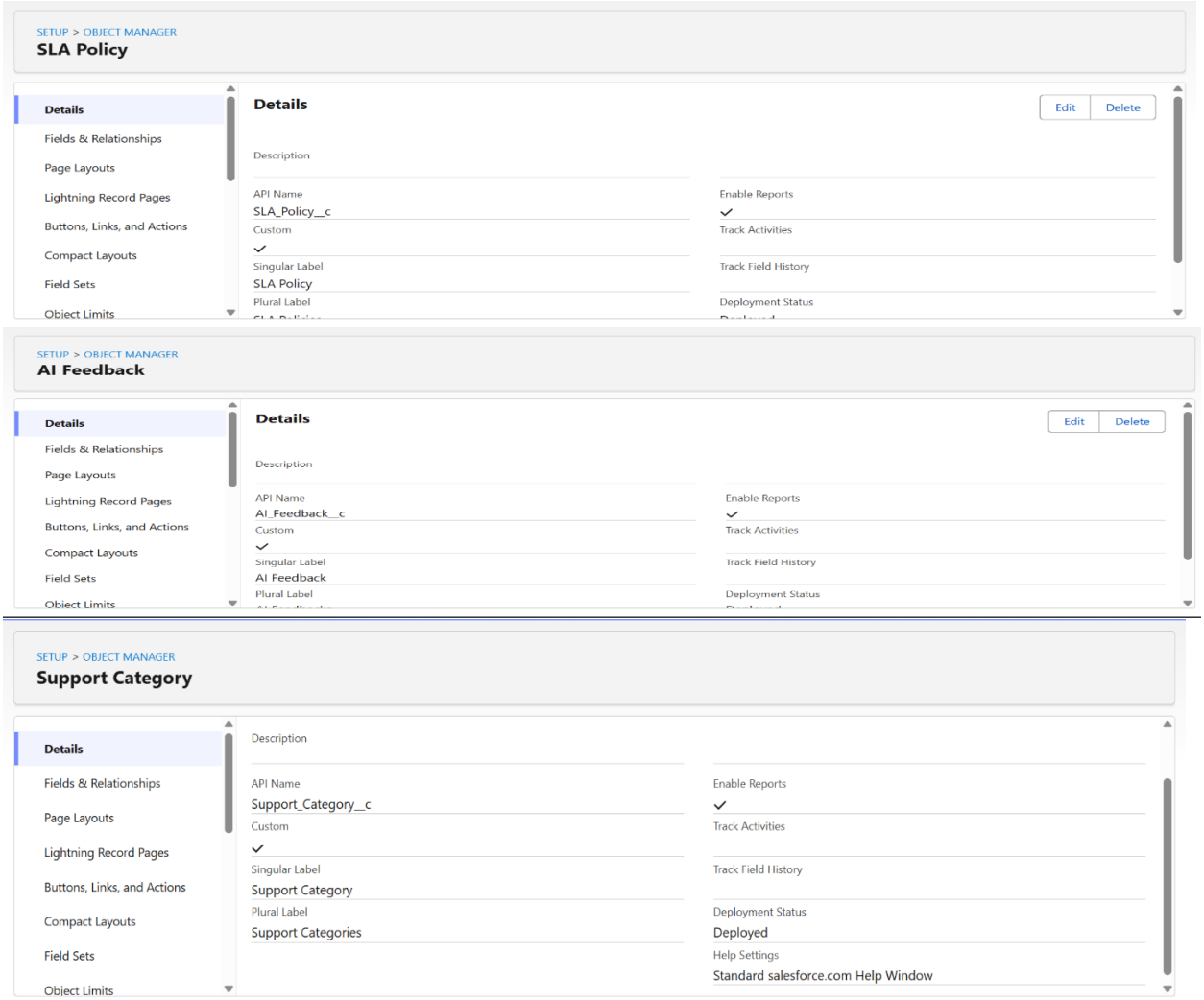


Figure 4: Custom Objects Created for the System

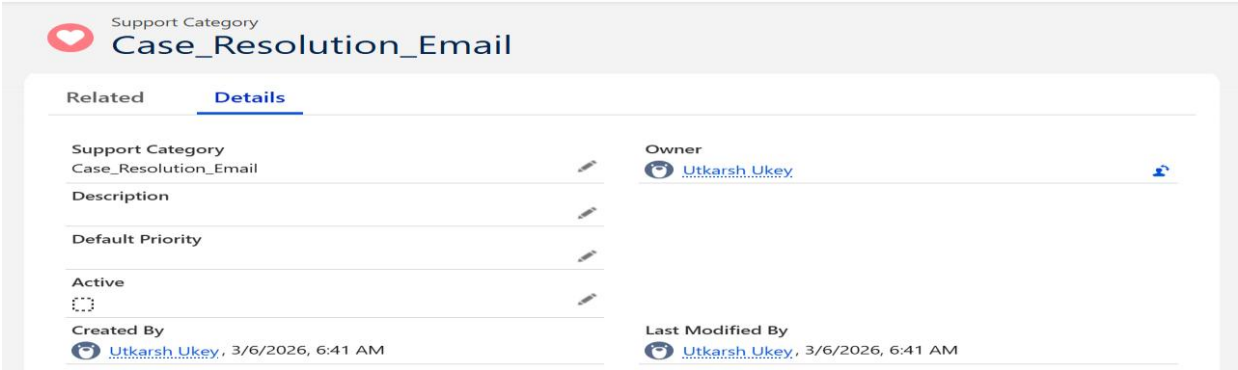


Figure 5: Support Category Custom Object

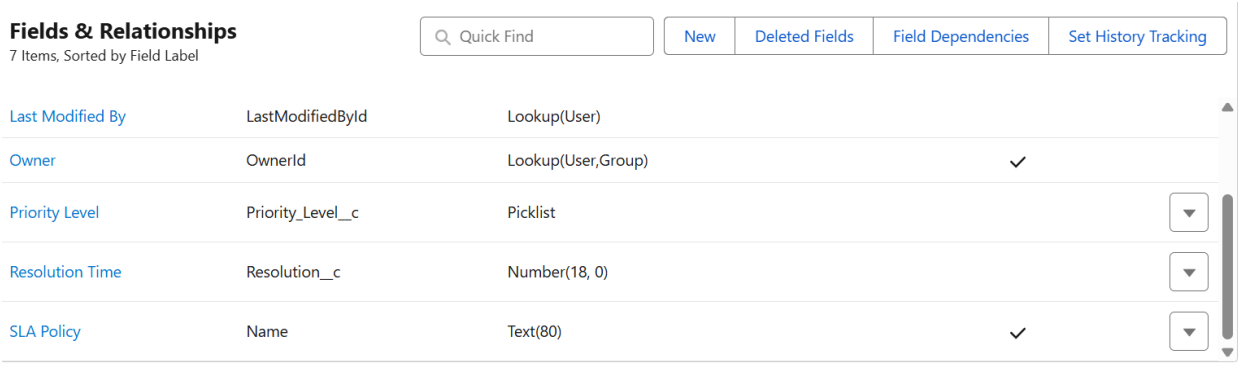


Figure 6: SLA Policy Object for Service Level Agreements

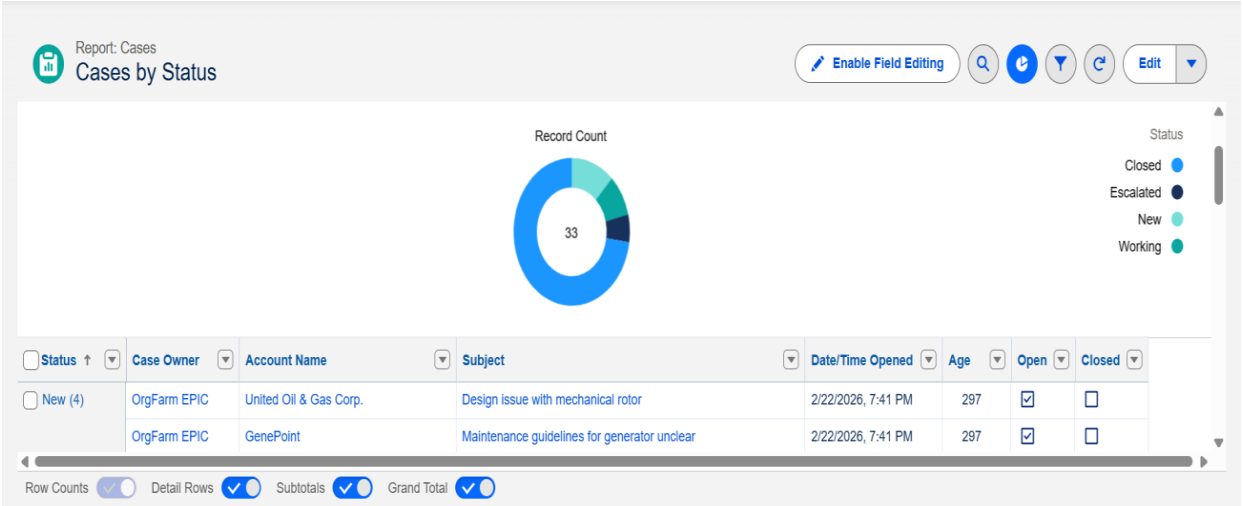
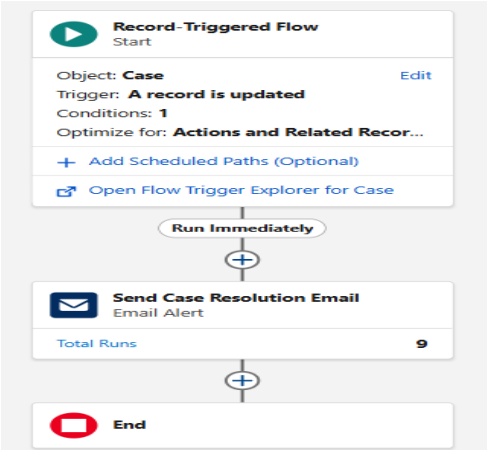


Figure 7:Case Status Report



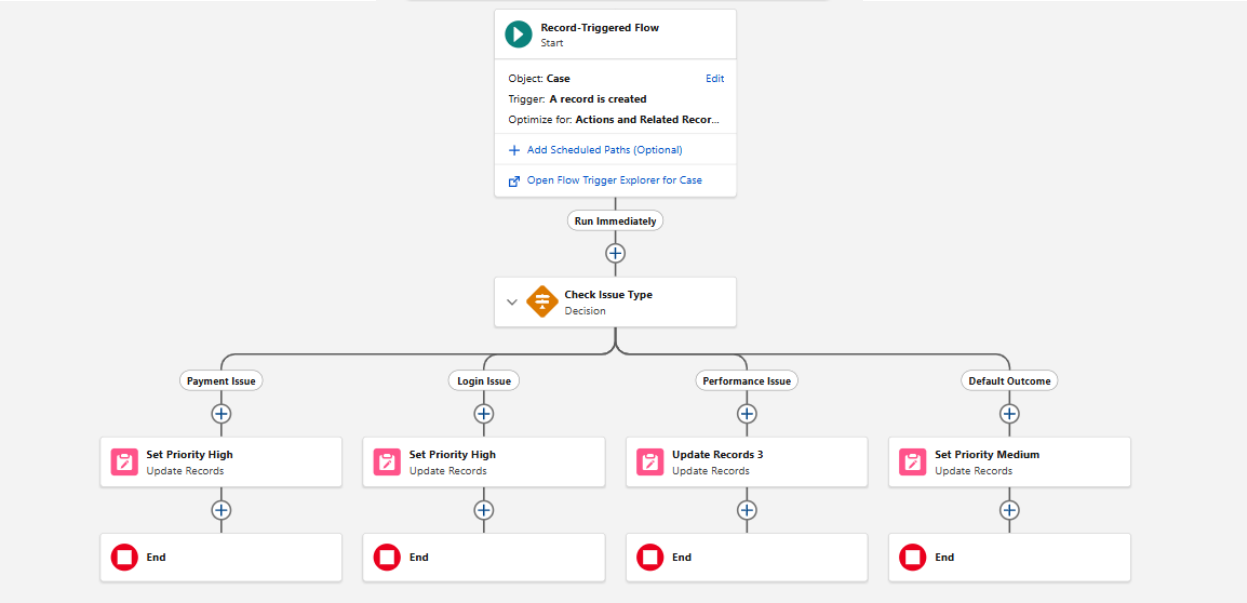
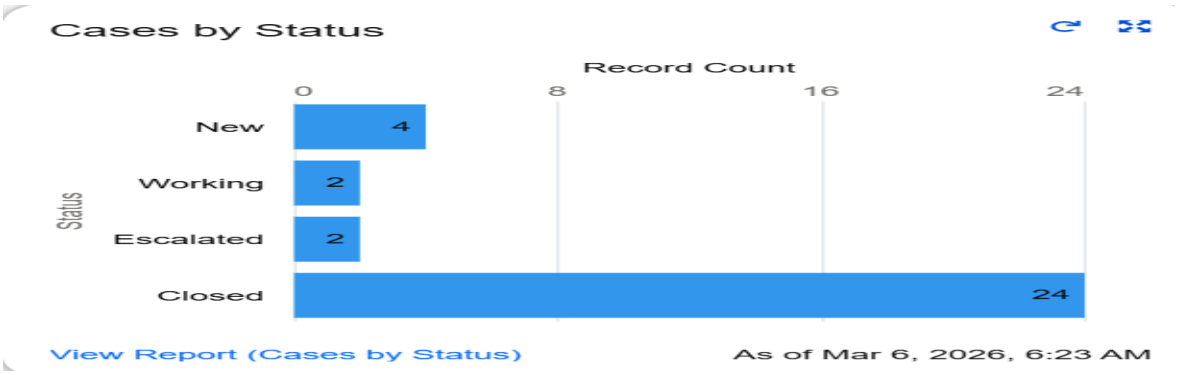
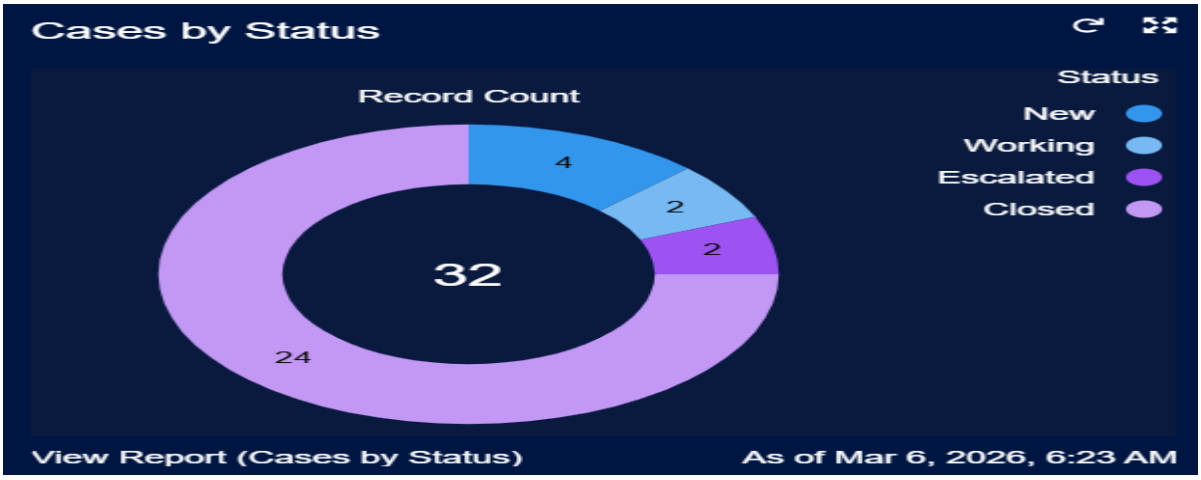


Figure 8 and Figure 8.1 : Case Resolution Email Flow and Case Priority Automation with SLA Due Date Calculation





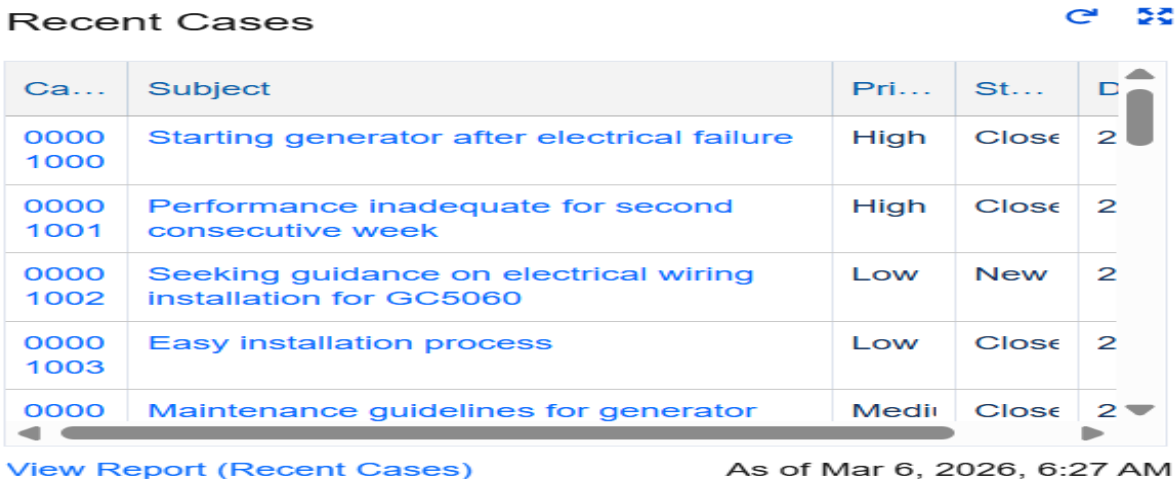


Figure 9: Customer Support Performance Dashboard

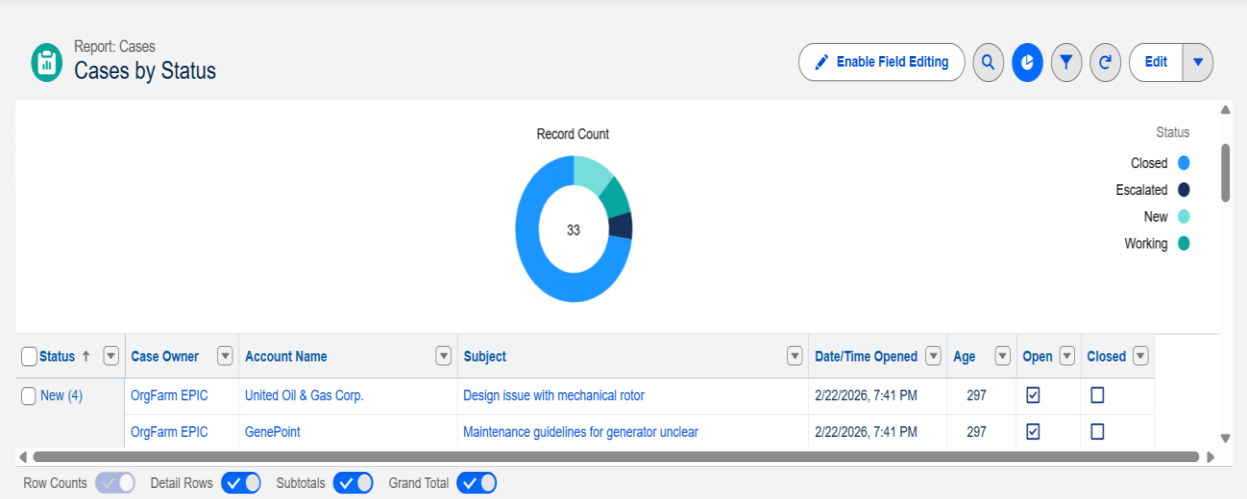


Figure 10: Example of Case Creation in the System

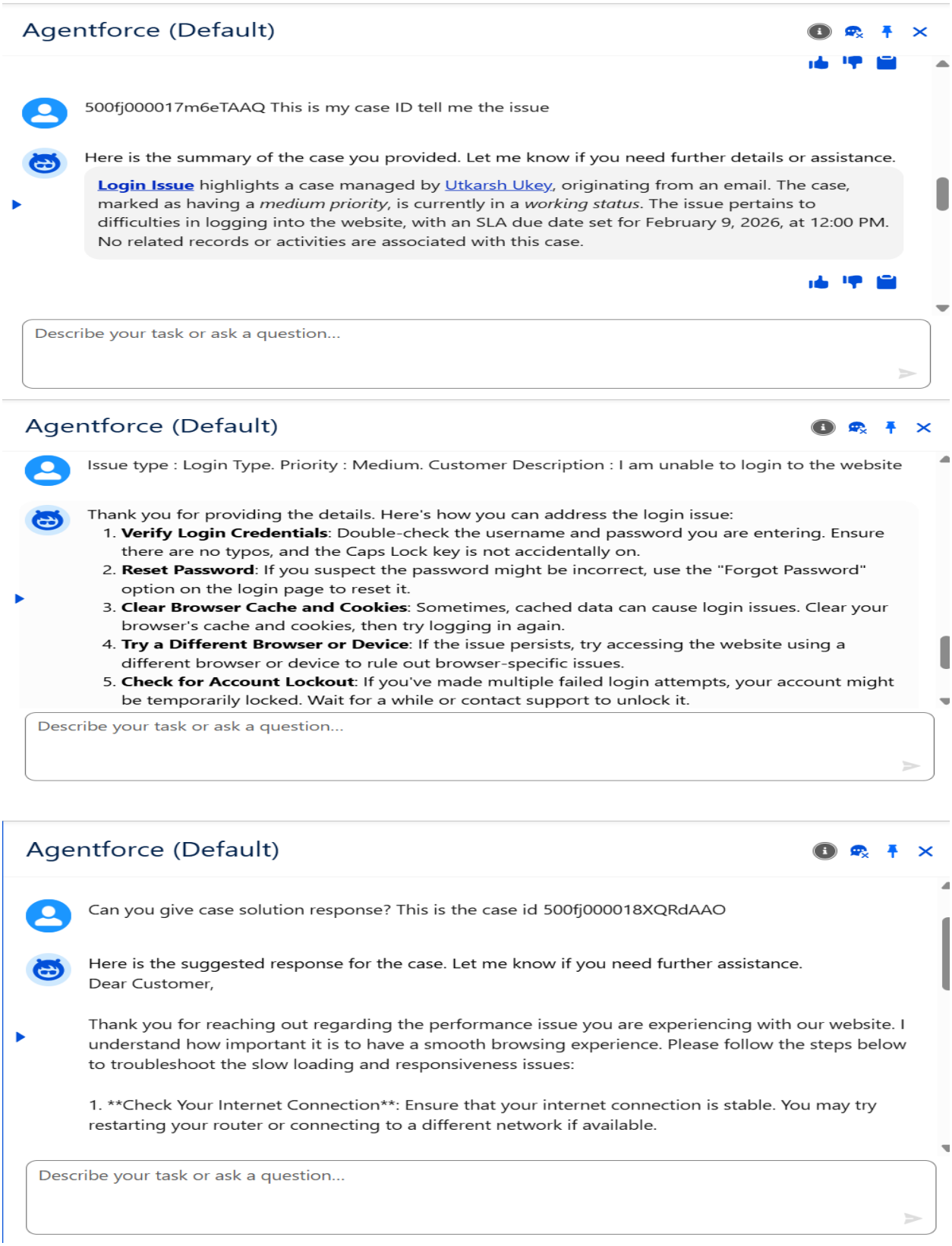


Figure 11: AI-powered assistance generating recommended actions for the support agent.